INAES Software User Manual

Comprehensive Guide: Installation, Workflows, Panel Tutorials, Theory, Export, and Troubleshooting

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1. Purpose and Scope

This manual explains how to use INAES as the current native desktop application for ice nucleation and freezing-assay analysis. It is written for operational use: what to load, what to click, what the software computes, how to export results, and how to diagnose common failures.

The document covers the PySide6 desktop software only. It does not describe the retired web/Dash interface. The active user-facing entry point is the INAES macOS application bundle or, for source execution, desktop_migration_pyside6/run_desktop.py.

- Covered panels: Data Upload nM, Freezing Curves, Compare Samples FC, Frozen Fraction, Kneepoint, Boxplots, and Correlations.
- Covered infrastructure: startup splash/preflight, session persistence, plot export, kneepoint report export, Frozen Fraction and Freezing Curves batch reports, and cancellable long-running tasks.
- Scientific caution: INAES preserves the implemented analytical logic; changes to normalization, smoothing, or breakpoint settings should be treated as analysis decisions and recorded with exported results.

2. Software Overview

INAES is a native PySide6 desktop application for transforming frozen-fraction assay data into normalized nucleation metrics, plotting freezing curves, comparing samples, identifying kneepoints, and linking curve-derived nM metrics to metadata.

- Data enter through the Data Upload nM tab as either already analyzed curves or RAW frozen-fraction tables.
- Column mapping standardizes user files into canonical fields used by all analysis panels.
- Interactive plots are Plotly figures embedded in the desktop UI through QWebEngineView.

- Heavy tasks such as RAW analysis, boxplot/correlation preparation, kneepoint analysis, and report generation run through background workers with progress and cancel controls.
- The main run buttons and cancel buttons remain visible in analysis tabs while the left settings panel scrolls.

The interface is organized around explicit execution. Changing filters or settings does not silently recompute every plot; the user clicks the panel run/update button to produce the current result.

3. System Requirements

- Recommended operating system for the current clickable bundle: macOS on Apple Silicon.
- For source execution: Python 3.13 virtual environment with PySide6, pandas, numpy, scipy, statsmodels, scikit-learn, pwlf, plotly, and qt-themes.
- Static plot export and kneepoint report export require Kaleido. If Kaleido is unavailable, interactive viewing may still work, but SVG/PDF/PNG generation can fail.
- The app writes session data to ~/.inaes_desktop/session_state_v1.json and launcher logs to ~/Library/Logs/INAES/inaes_desktop_app.log when opened from the macOS app bundle.
- The macOS launcher uses a runtime copy under ~/Library/Application Support/INAES/runtime to avoid Desktop privacy restrictions.

4. Installation Tutorial

Tutorial 0A - Launch on macOS from the app icon

1 Locate INAES.app on the Desktop or in Applications.

2 Double-click INAES.app.

3 The splash screen appears, shows startup checks, and then opens the main INAES window.

4 If macOS blocks the first launch, right-click INAES.app and choose Open.

5 If the app appears to do nothing, inspect ~/Library/Logs/INAES/inaes_desktop_app.log for launcher errors.

Tutorial 0B - Regenerate the local macOS launcher

Development builds can regenerate the clickable app without freezing the scientific code. The launcher copies the project and virtual environment into Application Support and points the app bundle at that runtime.

1 Open Terminal.

2 Run /Users/tommasolamesta/Desktop/INAES_python/desktop_migration_pyside6/scripts/create_macos_launcher_app.sh "/Users/tommasolamesta/Desktop/INAES.app".

3 Open the regenerated INAES.app.

4 Keep the source command as a fallback while testing new releases.

Tutorial 0C - Source execution fallback

1 cd /Users/tommasolamesta/Desktop/INAES_python

2 source .venv/bin/activate

3 python desktop_migration_pyside6/run_desktop.py

Tutorial 0D - Splash and startup preflight

At startup the app creates a QApplication, applies the built-in theme, shows a splash screen, and performs lightweight checks before importing the full main window. Current checks include application files, Qt runtime, pandas, numpy, plotly, scipy, scikit-learn, pwlf, session folder writability, temporary workspace writability, and optional Kaleido availability.

For emergency debugging, INAES_DISABLE_SPLASH=1 can be set in the environment before launching from source. This bypasses the splash layer without changing the scientific code.

5. Core Data Model

5.1 Curves file expected semantics

All downstream analysis panels expect a standardized curves table. The most important canonical fields are Sample, Size, Freezing.temperature, nm, Control, Dilution.factor, Location, FF, Sample.name, and Sample_ID.

- Sample identifies the biological/environmental sample after parsing and mapping.
- Size is commonly b_5_m or b_02_m, but the software preserves other valid labels.
- Freezing.temperature is numeric temperature in degrees Celsius.
- nm is the normalized nucleation metric selected by the workflow.
- Control is normalized toward Yes/No semantics and is used to include or exclude controls in curve panels.
- Dilution.factor is numeric and canonicalized for filtering and plotting.
- Sample_ID is a composite curve identifier used by curve grouping and plotting logic.

5.2 RAW file expected semantics

Typical RAW input contains columns such as Content, Freeze Temp, and Frozen Fraction. The UI can auto-map these to Sample.name, Freezing.temperature, and FF, but explicit mapping is recommended when column names differ.

- Well Name is not required for the calculations.
- Sample names can be parsed for size tokens and trailing dilution tokens.
- Optional RAW mappings include Size, Location, Control, and Dilution.factor.

- Rows missing required Sample, temperature, or FF values are dropped with a warning. Rows with FF >= 1 are protected by setting FF, nm, and Freezing.temperature to NA in the analyzed output.

5.3 Metadata file semantics

Metadata must include the Sample column used for merging with curve-derived metrics. Other metadata columns remain available as grouping variables for Boxplots and X/Y candidates for Correlations.

6. Mathematical and Statistical Logic

6.1 Vali transform

Most RAW normalization methods start from Lambda(T) = -ln(1-FF). The app treats FF numerically, validates required parameters, and propagates warnings when rows are removed or protected.

6.2 Normalization methods implemented

- Mass-normalized nM: nM(T) = -(1/V) ln(1-FF) * (d / c_m), where V is droplet volume in mL, d is Dilution.factor, and c_m is mass concentration in g/mL. Units: g^-1.
- Per liquid volume K(T): K(T) = -(1/V) ln(1-FF). Units: mL^-1.
- Standard RAW nM: nM(T) = -ln(1-FF)/(30*10^-3) * (Water_volume/Soil_mass) * Dilution.factor. This preserves the historical 30 uL constant for this method.
- Surface-site density direct: n_s(T) = -ln(1-FF) / A_per_droplet. Units: m^-2.
- Surface-site density from BET: n_s,BET(T) = nM(T) / theta, where theta is BET area. Units: m^-2.
- Per-cell cumulative metric: n_cell(T) = -(1/V) ln(1-FF) * (d / c_cells). V is the selected droplet volume in mL; for Micro-PINGUIN defaults V = 0.03 mL. Units: cell^-1.
- Fungal biomass N_m: N_m(T) = [-ln(1-FF) / Vdrop] * (Vwash / m) * d, where Vwash is the volume of the mother suspension in mL, m is the fresh mycelium mass in g, and d is the dilution factor of the droplet series relative to the mother suspension (TINA scheme, Kunert et al.). Units: g^-1 of mycelium.
- Purified protein N_n (ice nucleation site density per molecule): N_n(T) = -ln(1-FF) * d / N_drop, where N_drop is the number of protein molecules in one droplet of the undiluted solution. With the mass input, N_drop = (c_m / M) * Vdrop * N_A, with c_m in mg/mL and M in kDa (converted to g/mL and g/mol), so that N_n(T) = -ln(1-FF) * d * M / (c_m * Vdrop * N_A). With the molar input, N_drop = c_mol * Vdrop * N_A, with c_mol in uM. N_A = 6.02214076 x 10^23 mol^-1. Units: molecule^-1. Reference: Hartmann et al. (2022), Front. Microbiol. 13:872306, doi:10.3389/fmicb.2022.872306.
- Air wash-off: N_INP = -ln(1-FF) * (Vwash / (Vdrop * x * Vs)). Units: L^-1.

- Air drop-on: N_INP = -ln(1-FF) * (A_filter / (alpha * Vs)). Units: L^-1.
- Custom dose-per-droplet: nX(T) = -ln(1-FF) / X. Units: X^-1.

6.3 Normalization for purified proteins

For purified or recombinant ice-nucleating proteins the biologically meaningful dose is the number of protein molecules in each droplet, so INAES reports the ice nucleation site density per molecule, N_n(T), as in Hartmann et al. (2022). Per-mass values (g^-1 of protein) remain available through the mass-normalized nM method; the two are related by N_n = nM * M / N_A.

- c_m, c_mol and M refer to the undiluted protein solution; the dilution factor d of each droplet series is applied by INAES.
- M is the molar mass of the construct as used for the concentration (usually the monomer). Ice-nucleating proteins aggregate, so N_n is read as active sites per molecule, not as the fraction of molecules that are active. Values are typically very small (for example 10^-15 to 10^-6 per molecule) and are best viewed on a log axis.
- The RAW status line reports the number of molecules per droplet at d = 1, which is a quick check of the entered concentration.

6.4 nM10 / nM15 computation from curves

metadata_with_nm is computed by LOESS prediction at target temperatures. Defaults are -10 C and -15 C, with optional custom targets from the Data Upload nM settings.

- Coerce nm and Freezing.temperature to numeric.
- Drop invalid rows and trim nm to the 5-95% interval per sample group.
- Fit LOESS per Sample and Size subset.
- Predict metrics at target temperatures and merge them into metadata as nM10_b5, nM15_b5, nM10_b02, nM15_b02, plus custom target columns when configured.
- Negative predictions are converted to NA.

6.5 Freezing Curves mean +/- 95% CI summary logic

The Freezing Curves panel produces both a primary scatter plot and a mean +/- 95% CI summary. The mean summary groups curves, interpolates them onto a common temperature grid, computes summary behavior in log space, and avoids bridging unsupported temperature gaps.

6.6 Kneepoint detection logic

Kneepoint analysis works on log10(nm) versus temperature. The data are filtered by sample, size, dilution, and optional temperature range, then smoothed with a spline and fitted with piecewise linear segments. Candidate breakpoints are classified by slope-state transitions, and the relevant positive-to-negative transitions are kept as kneepoints.

Diagnostics include raw and smoothed point counts, selected breakpoints, non-knee breakpoints, linear versus piecewise residual sum of squares, bootstrap success, and cross-validation metrics.

7. Global UI Concepts

- Each panel has an explicit run/update button and a cancel button where the task can be long-running.
- The primary action row is sticky in analysis tabs: it remains visible while scrolling panel options.
- Data Upload retains the table preview area; heavy filtered preview tables are not shown in the analysis tabs.
- Universal style controls include palette, background, grid visibility, title/subtitle or title prefix, marker borders, and Y-axis tick style where relevant.
- Export controls are panel-specific and accept format, width, height, scale, and filename stem.
- The nM axis label is propagated through the shared app state and can be inferred from RAW method metadata or manually overridden for analyzed-file workflows.

8. Tutorial 1 - Startup Workflow Selection

1 Open INAES.

2 Stay in the Data Upload nM tab.

3 Choose the workflow mode: Upload analyzed file, Analyze RAW -> nM, or Update existing analyzed with RAW.

4 Review the hint text under the mode selector; the left-side controls adapt to the selected workflow.

5 Load or map data as required before moving to analysis panels.

The selected mode is saved in session settings. Datasets themselves are not automatically loaded on startup.

9. Tutorial 2 - Data Upload nM Panel

9.1 Path A: Upload analyzed file

1 Click Load analyzed file.

2 Load optional metadata with Load metadata file.

3 Use Curves Standardization to map Sample, Size, Location, Freezing.temperature, nm, Control, Dilution.factor, and FF.

4 Use Auto-map if column names are close to expected aliases; otherwise choose mappings explicitly.

5 Apply standardization, then inspect the Data Upload preview and status line.

6 Set nM axis label behavior and LOESS/custom-temperature settings if metadata_with_nm will be used.

9.2 Path B: Analyze RAW and compute nM

1 Select Analyze RAW -> nM.

2 Click Load RAW file.

3 Map Sample.name, Freezing.temperature, and FF; map optional Size, Location, Control, and Dilution.factor if present.

4 Choose grouping behavior, auto dilution parsing, and control keyword detection.

5 Select Platform setup. Micro-PINGUIN defaults use 30 uL droplet volume; Custom setup allows a user-selected droplet volume and method parameters.

6 Choose the RAW normalization method and fill all required method parameters. Only the fields needed by the selected method are shown. For Purified protein N_n, first choose Protein amount input: Mass concentration + molar mass (c_m in mg/mL and M in kDa) or Molar concentration (c_mol in uM). For Fungal biomass N_m, enter the suspension volume V_wash and the fresh mycelium mass m.

7 Click Analyze RAW (Vali). Use Cancel if needed.

8 Inspect the RAW status and preview.

9 Click Use analyzed file to send the analyzed table to the active curves source, or Save analyzed CSV for external archiving.

9.3 Path C: Update existing analyzed file with RAW

1 Select Update existing analyzed with RAW.

2 Load the new RAW file and the previous analyzed file.

3 Run RAW analysis as in Path B.

4 Map analyzed RAW output fields into the previous analyzed schema using the merge mapping controls.

5 Click Merge into previous analyzed. Use Cancel merge if needed.

6 Click Use merged file to activate the merged table in the app, or Save merged CSV to preserve it.

Update mode is schema-preserving. The merged output keeps the previous file column order and excludes unmapped extra RAW fields by design.

10. Tutorial 3 - Freezing Curves Panel

1 Open Freezing Curves.

2 Select Size, Dilution.factor, and Location filters.

3 Choose whether controls are included.

4 Adjust palette, point border, background, grid, Y tick style, title, and subtitle.

5 Configure mean +/- 95% CI settings such as grouping, temperature step, smoothing, and CI method where available.

6 Click Update Curves. Cancel remains available during computation.

7 Use the separate export boxes for the main scatter plot and the mean curve plot.

8 To save one panel per sample, use Create Report (batch); see Tutorial 7B.

The primary plot displays filtered nm versus Freezing.temperature on a log-scaled Y axis. The summary plot displays grouped mean curves and confidence bands.

11. Tutorial 4 - Compare Samples FC Panel

1 Open Compare Samples FC.

2 Select up to the UI-supported set of samples for targeted comparison.

3 Select Size and Dilution.factor filters.

4 Adjust sample shapes, palette, point border, background, grid, Y tick style, and title prefix.

5 Click Update Curves. Use Cancel if the dataset is large.

6 Export the current comparison plot if needed.

This panel is intended for focused visual comparison under consistent filtering and trimming. It excludes control rows unless the implemented filter path includes them explicitly.

12. Tutorial 5 - Frozen Fraction Panel

1 Open Frozen Fraction.

2 Select one or more samples. When data are loaded, only the first sample is selected; the plot shows exactly the selected samples.

3 Select Size and Dilution.factor filters.

4 Enable or disable control point display.

5 Choose the Color mode (see below), then customize palette, sample/control symbols, marker border, background, grid, and title.

6 Click Update Frozen Fraction. Use Cancel if needed.

7 Export the FF plot in SVG, PDF, or PNG.

8 To compare many samples, use Create Report (batch), which draws one panel per sample (Tutorial 7B).

The Frozen Fraction plot uses FF against temperature and canonicalizes dilution values so equivalent numeric labels filter consistently.

Color modes:

- Auto (default): one distinct color per dilution when one sample is shown; one hue per sample with shades per dilution when several samples are shown.
- Distinct color per dilution: one palette color per dilution, all samples pooled.
- Sample hue, shades per dilution: each sample gets its own palette hue; dilutions are shades of that hue, darkest for the most concentrated (lowest dilution factor). With a single sample this is a one-color gradient. The legend is grouped by sample.
- Solid color per sample: one palette color per sample.
- One color for all samples, shades per dilution: the first palette color for every sample, shaded by dilution.
- One solid color for all samples: the first palette color for every point.

13. Tutorial 6 - Kneepoint Panel

1 Open Kneepoint.

2 Select one Sample, one Size, and one or more Dilution.factor values.

3 Optionally restrict the temperature range.

4 Set spar-like smoothing. Higher values produce stronger smoothing; zero approaches interpolation.

5 Set requested breakpoint behavior and segment selection mode if exposed in the UI.

6 Tune flat and rise quantiles if candidate classification needs adjustment.

7 Adjust point size, line width, grid, background, and title styling.

8 Click Analyze Kneepoint or Run Kneepoint. Cancel remains available during computation.

9 Inspect the plot, result table, and status diagnostics.

Treat kneepoints as analytical candidates, not automatic truth. Always inspect whether the selected transition is physically meaningful for the sample and temperature range.

14. Tutorial 7 - Kneepoint Report Generation

1 Complete or configure Kneepoint settings for Size, Dilution.factor, temperature range, smoothing, and breakpoint options.

2 In the report section, select report samples or use Select all.

3 Create the report preview. The preview step computes per-sample kneepoint results and builds editable sample report items.

4 Review skipped samples and per-sample notes.

5 Edit sample report cards if needed through the available report editor controls.

6 Export or download the report ZIP. Use Cancel while the report worker is running if needed.

The exported ZIP contains summary CSV files and static figures such as per-sample plots, an all-samples grid, and a full report figure. SVG/PDF generation requires Kaleido. Export status includes packaged files, skipped samples, and export warnings.

15. Tutorial 7B - Frozen Fraction and Freezing Curves Reports

Both the Frozen Fraction and the Freezing Curves panels contain a Create Report (batch) box that saves one figure with one panel per sample, in the style of a faceted grid.

1. Set the panel filters first (Size, Dilution.factor, controls, and Location for Freezing Curves) and the plot style (palette, point size, marker border, background, title). The report uses the same filtered points as the panel.
2. In Create Report (batch), select the report samples or use Select All.
3. Choose Panel colors, using the same color modes listed in Tutorial 5, including One color for all samples.
4. Choose Columns (Auto or 1 to 8) and whether all panels share the same axis ranges (Same axis ranges in all panels, on by default).
5. Choose the formats (PNG, SVG, PDF; all are selected by default), the save folder, and the file prefix.
6. Click Build Preview. The preview appears below the panel plots and is interactive.
7. Click Save Report. The saved files are exactly the previewed figure.

- One panel is drawn per sample. When the selection contains more than one Size, panels are split as sample | size.
- Files are named prefix_YYYYMMDD_HHMMSS.png/.svg/.pdf, so earlier reports are never overwritten.
- Each panel is 360 x 280 px in the saved figure; PNG files are written at 2x resolution. SVG and PDF stay vector.
- The Freezing Curves report uses a log Y axis labelled with the active nm axis label; the Frozen Fraction report uses FF from 0 to 1.
- Changing the sample selection clears the preview; build it again before saving. Report export requires Kaleido.

16. Tutorial 8 - Boxplots Panel

1. Load curves and metadata in Data Upload nM.
2. Wait for metadata_with_nm options to refresh; this can run in the background.
3. Open Boxplots.
4. Choose the response variable, usually nM10 or nM15 for a specific size suffix.

5 Choose the Size facet and Group-by metadata variable.

6 If Group-by is continuous numeric, enable binning and choose either number of ranges or fixed range width.

7 Adjust scale, jitter visibility, palette, point size, marker border, title, grid, and background.

8 Click Update Boxplot. Use Cancel for large datasets.

9 Export the current plot if needed.

Boxplots are based on metadata merged with curve-derived nM metrics. If expected response variables are missing, return to Data Upload and verify metadata, LOESS settings, and sufficient curve points per sample-size group.

17. Tutorial 9 - Correlations Panel

1 Load curves and metadata in Data Upload nM.

2 Open Correlations.

3 Choose correlation method: Spearman, Pearson, Quadratic Fit, or GAM.

4 Choose X and Y variables. Y may be size-aware nM10/nM15 or a generic numeric metadata field.

5 Optionally filter by Location.

6 Customize location shapes, point size, marker border, palette, title prefix, background, grid, and Y tick style.

7 Click Run Correlation. Use Cancel for large datasets.

8 Review fit line/curve, CI band when available, and diagnostics in labels/status.

9 Export the current correlation plot if needed.

For nM-like Y values, the plot uses positive values for log-scale behavior. Spearman and Pearson provide correlation summaries; Quadratic and GAM provide fitted curves when enough valid data are available.

18. Export Tutorial (All Plot Panels)

1 Run the target panel so a current Plotly figure exists.

2 In the panel export box, choose SVG, PDF, or PNG.

3 Set width and height in pixels. For PDF, scale is treated conservatively by the exporter.

4 Set the filename stem.

5 Click Export plot.

6 Choose the save location in the file dialog.

SVG is preferred for publication-quality vector editing. PNG is convenient for slides and quick sharing. PDF is useful for fixed-layout reports. All static export paths rely on Plotly/Kaleido.

19. Column Mapping and Parsing Rules (Reference)

- Curves standardization maps arbitrary column names to canonical fields using explicit UI selections or aliases.
- RAW auto-mapping recognizes common names such as Content, Freeze Temp, Frozen Fraction, sample, temperature, control, blank, dilution, site, and size.
- Trailing dilution parsing recognizes a final sample-name token that is a power of ten (10, 100, 1000, ... with no upper limit, for example C6_1000000), with an optional x suffix such as 100x.
- Size parsing recognizes common INAES size tokens such as b_5_m and b_02_m; unknown values are preserved when explicitly mapped.
- Control detection uses user-configured keywords across sample names and mapped control fields.
- Leading control keyword prefixes can be stripped from sample names for grouping consistency while the Control column remains Yes/No-like.

20. Troubleshooting (Comprehensive)

20.1 App does not open from icon

- Check ~/Library/Logs/INAES/inaes_desktop_app.log.
- If the log mentions pyvenv.cfg permission errors, regenerate the launcher so it uses the Application Support runtime.
- If the log mentions incompatible architecture for numpy or scipy, confirm the launcher uses /usr/bin/arch -arm64.
- If no log appears, try right-click Open to satisfy Gatekeeper.

20.2 Splash appears but main window does not complete

- Inspect the launcher log and the runtime log reported in the terminal or app output.
- Missing PySide6, pandas, numpy, plotly, scipy, scikit-learn, or pwlf indicates an incomplete runtime environment.
- Missing Kaleido affects export/report generation but should be treated as optional startup warning.

20.3 Freezing curve shows FF-like values instead of normalized metric

- Verify that RAW normalization method and required parameters are correct.
- For per-cell cumulative metric, confirm droplet volume and cell concentration are set. The implemented formula is -(1/V) ln(1-FF) * (d/c_cells).
- Confirm the active curves source is the analyzed RAW output, not the original RAW table.

- Inspect Data Upload preview for non-null nm values and expected normalization_label/normalization_units columns.

20.4 Export fails

- Install or repair Kaleido in the active runtime.
- Try SVG first, then PNG or PDF.
- Reduce width/height/scale if the generated image is very large.
- Check write permissions for the destination folder.

20.5 Boxplots or correlations show missing variables

- Confirm metadata was loaded and includes Sample.
- Confirm curves contain enough valid numeric nm/temperature rows per Sample and Size.
- Review LOESS span/min-points settings in Data Upload nM.
- Wait for background option refresh to finish after changing datasets.

21. Performance and Reliability Tips

- Filter early by sample, size, dilution, and location for interactive speed.
- Use Cancel when a run was started with incorrect settings; then adjust and rerun.
- Keep numeric columns clean: decimal dot, finite values, and no mixed text suffixes in measurement fields.
- Record RAW method parameters with exported results, especially droplet volume, concentration, dilution behavior, and temperature range.
- Avoid repeatedly switching very large datasets while background option refreshes are running.
- Prefer SVG export for final figure editing; use moderate PNG scale values for quick review images.

22. Validation Checklist Before Sharing Results

1. Confirm the workflow mode matches the actual data source.
2. Confirm mapping for Sample, Size, Freezing.temperature, nm or FF, Control, Dilution.factor, and Location.
3. Confirm active curves source after RAW analysis or RAW update.
4. Confirm nM axis label and units match the selected normalization method.
5. For per-cell metric, verify droplet volume and cells/mL before interpreting curves.
6. For Purified protein N_n, verify the protein amount input (c_m and M, or c_mol), that it refers to the undiluted solution, and the molecules-per-droplet value in the RAW status.

7 For Fungal biomass N_m, verify V_wash, the fresh mycelium mass m, and the dilution factors relative to the mother suspension.

8 For Freezing Curves and Compare Samples, verify controls are included or excluded intentionally.

9 For Kneepoint, verify temperature range, smoothing, and candidate breakpoints visually.

10 For Boxplots and Correlations, verify metadata merge and response variable semantics.

11 Archive exported figures/report ZIP together with the input data and settings used.

23. Glossary

- FF: Frozen Fraction, typically between 0 and 1.
- Lambda(T): Vali transform, -ln(1-FF).
- nM: Generic normalized nucleation metric; exact units depend on the selected method.
- n_cell: Per-cell cumulative metric in cell^-1.
- N_m: Fungal biomass metric, ice nucleation sites per gram of fresh mycelium (g^-1).
- N_n: Purified-protein metric, ice nucleation sites per protein molecule (molecule^-1).
- N0: Number of droplets or assay units used by RAW Vali handling.
- LOESS: Local regression smoother used for nM prediction at target temperatures.
- Kneepoint: Breakpoint on smoothed log10(nm)-temperature behavior retained after slope-transition filtering.
- Kaleido: Plotly static export engine used for SVG/PDF/PNG export.
- Sample_ID: Composite identifier used for curve grouping.

24. Appendix - Practical Mapping Examples

Example A: RAW default columns. Map Content to Sample.name, Freeze Temp to Freezing.temperature, and Frozen Fraction to FF. Use Micro-PINGUIN defaults when the assay uses 30 uL droplets.

Example B: analyzed file with non-standard labels. Map the measured metric column to nm, map the temperature column to Freezing.temperature, map sample/grouping columns explicitly, and then run Curves Standardization.

Example C: no Size or Location column. Disable the corresponding grouping or provide manual/single-group labels so downstream panels receive stable values.

Example D: per-cell RAW analysis. Select Per-cell cumulative metric, set Cell concentration in cells/mL, verify Droplet volume in uL, and confirm the output axis label becomes n_cell (cell^-1).

Example E: updating a previous analyzed file. Load the previous analyzed table, analyze the new RAW file, map the RAW analyzed fields into the previous schema, merge, inspect status, then

save the updated CSV.

Example F: purified protein. Select Purified protein N_n (per molecule). For a 1 mg/mL stock of a 100 kDa construct, choose Mass concentration + molar mass with c_m = 1 and M = 100, or equivalently Molar concentration with c_mol = 10 uM. With 30 uL droplets the RAW status reports about 1.81 x 10^14 molecules per droplet at d = 1, and the axis label becomes N_n (molecule^-1).

Example G: fungal biomass. Select Fungal biomass N_m, enter the suspension volume V_wash and the fresh mycelium mass m, and make sure the dilution factors are relative to the mother suspension. The axis label becomes N_m (g^-1).

End of manual.